Jaemoo Choi

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Research Interests

- **Generative AI**: Diffusion Models, Video/World Models, Distillation, LLM fine-tuning
- **AI for Science**: Boltzmann distribution sampling
- **Optimal Control**: Optimal Transport, Schrödinger Bridge, Optimization

Education & Positions

09/2024 – present **Georgia Institute of Technology** Atlanta, GA
Postdoctoral Associate
Advisor: Yongxin Chen

03/2018 – 08/2024 **Seoul National University (SNU)** Seoul, Korea
Ph.D. *in Mathematical Sciences*
Advisor: Myungjoo Kang

03/2014 – 02/2018 **Seoul National University (SNU)** Seoul, Korea
B.S. *in Mathematics Education*

Selected Publications

[1] **Rethinking the Design Space of Reinforcement Learning for Diffusion Models: On the Importance of Likelihood Estimation Beyond Loss Design**
ICML, 2026
In reward-based fine-tuning for diffusion models, accurate likelihood estimation is more important than the precise choice of RL loss. ELBO-based estimators make fine-tuning substantially efficient and stable.

[2] **Efficient Adjoint Matching for Fine-tuning Diffusion Models**
NeurIPS, 2026 [Spotlight, 1.3%]
We make adjoint matching, a principled stochastic-optimal-control approach to reward-based fine-tuning of diffusion models, substantially more efficient while preserving its stability.

[3] **QUATRO: Query-Adaptive Trust Region Policy Optimization for LLM Fine-tuning**
ICML, 2026
This work gives a theoretically grounded view of GRPO-style LLM fine-tuning by deriving an exact query-wise trust-region bound objective. This leads to more stable updates and helps reduce mode collapse.

[4] **Adjoint Schrödinger Bridge Sampler**
NeurIPS, 2025 [Oral, 0.3%]
We introduce an efficient first-order method for reward-based diffusion sampler by extending adjoint matching. The method improves exploration and mode coverage in difficult sampling problems.

[5] **Discrete Adjoint Schrödinger Bridge Sampler**
ICML, 2026
We develop a discrete-state counterpart of first-order reward-based diffusion sampler methods such as adjoint matching and adjoint Schrödinger bridge sampler. This makes principled sampling possible even in discrete state spaces where standard gradients do not exist.

Industry Collaboration Projects

Scalable Reinforcement Learning for Diffusion Models **NVIDIA** · 2026
- We developed a simple trajectory-free method that incorporates reward feedback into velocity-matching.
- Developed an pipeline for standard video diffusion fine-tuning.

Diffusion Samplers with Force Fields and Energy Functions **FAIR, Meta** · 2025
- Developed a stable first-order framework for diffusion-based sampling using Schrödinger bridge methods.
- Studied sampling under force fields and energy-based objectives, with a focus on efficiency and stability.

Automatic Feature Extraction of Sensor Data — **Samsung Electronics** · 2020–2021

- Developed machine learning methods for automatic feature extraction from sensor data.
- Improved downstream analysis and decision-making for industrial sensing applications.

Automatic Real-World Video Enhancement — **four-by-four** · 2022–2023

- Served as Project Manager for a project on learning-based denoising and super-resolution for real-world video enhancement, targeting 4K and 8K video content from K-dramas and films.
- Addressed practical degradation sources by modeling diverse noise patterns and developing patch-based, multi-frame, and stitching-based restoration pipelines.

Honors & Awards

2026	Gold Reviewer for ICML 2026
2025	InnoCORE Postdoctoral Fellowship
2024	**Best Reviewer for NeurIPS 2024**
2024	Nurturing Next-generation Researchers Postdoctoral Fellowship ($ 45000)
2019	**Outstanding TA Awards**, SNU
2018 – 2022, 2024	Brain Korea Research Scholarship
2018	Beakwoon Academic Excellence Scholarship
2015 – 2017	Academic Excellence Scholarship, SNU
2014	Chungkwan Academic Exellence Scholarship, SNU

Publications

*Equal contribution *Core contributors †Equal advising ★First-author or corresponding-author paper

Conference Papers

NeurIPS 2026 (2 papers)

★ Efficient Adjoint Matching for Fine-tuning Diffusion Models [Spotlight, 1.3%]
Jeongwoo Shin*, Dongsoo Shin*, Yuchen Zhu, Wei Guo, Yongxin Chen, Joonseok Lee†, Jaewoong Choi†, **Jaemoo Choi**†

- Coarse-to-Fine Compositional Diffusion for Long-Horizon Planning
Byoungwoo Park, Utkarsh A. Mishra, **Jaemoo Choi**, Juho Lee, Yongxin Chen

ICML 2026 (7 papers)

★ Rethinking the Design Space of Reinforcement Learning for Diffusion Models: On the Importance of Likelihood Estimation Beyond Loss Design
Jaemoo Choi*, Yuchen Zhu*, Wei Guo, Petr Molodyk, Bo Yuan, Jinbin Bai, Yi Xin, Molei Tao, Yongxin Chen

★ QUATRO: Query-Adaptive Trust Region Policy Optimization for LLM Fine-tuning
Doyeon Lee, Eunyi Lyou, Hyunsoo Cho†, Sookyung Kim†, Joonseok Lee†, **Jaemoo Choi**†

- Enhancing Reasoning for Diffusion LLMs via Distribution Matching Policy Optimization [Spotlight, 2.0%]
Yuchen Zhu*, Wei Guo*, **Jaemoo Choi**, Petr Molodyk, Bo Yuan, Molei Tao, Yongxin Chen

★ Discrete Adjoint Schrödinger Bridge Sampler
Wei Guo*, Yuchen Zhu, Xiaochen Du, Juno Nam, Yongxin Chen†, Rafael Gómez-Bombarelli†, Guan-Horng Liu†, Molei Tao†, **Jaemoo Choi***

★ Generalized Schrödinger Bridge on Graphs
Panagiotis Theodoropoulos, Juno Nam, Evangelos Theodorou†, **Jaemoo Choi**†

★ Efficient Generative Modeling beyond Memoryless Diffusion via Adjoint Schrödinger Bridge Matching
Jeongwoo Shin, Jinhwan Sul, Jaewoong Choi†, Joonseok Lee†, **Jaemoo Choi**†

- MetaDNS: Enhancing Exploration in Discrete Neural Samplers via Metadynamics
Xiaochen Du, Juno Nam, **Jaemoo Choi**, Wei Guo, Sathya Edamadaka, Junyi Sha, Elton Pan, Yongxin Chen, Molei Tao, Rafael Gomez-Bombarelli

ICLR 2026 (1 paper)

★ Proximal Diffusion Neural Samplers
Wei Guo*, **Jaemoo Choi***, Yuchen Zhu, Molei Tao, Yongxin Chen

NeurIPS 2025 (3 papers)

★ Adjoint Schrödinger Bridge Sampler [Oral, 0.3%]
Guan-Horng Liu*, **Jaemoo Choi***, Yongxin Chen, Benjamin Kurt Miller, Ricky T. Q. Chen*

★ Non-equilibrium Annealed Adjoint Sampler
Jaemoo Choi[*], Yongxin Chen, Molei Tao, Guan-Horng Liu[*]

• MDNS: Masked Diffusion Neural Sampler via Stochastic Optimal Control
Yuchen Zhu[*], Wei Guo[*], **Jaemoo Choi**, Guan-Horng Liu, Yongxin Chen[†], Molei Tao[†]

ICML 2025
(2 papers)

★ Overcoming Spurious Solutions in Semi-Dual Neural Optimal Transport: A Smoothing Approach for Learning the Optimal Transport Plan
Jaemoo Choi, Jaewoong Choi[†], Dohyun Kwon[†]

• Unsupervised Point Cloud Completion through Unbalanced Optimal Transport
Taekyung Lee, **Jaemoo Choi**, Myungjoo Kang[†], Jaewoong Choi[†]

ICLR 2025
(2 papers)

★ Robust Barycenter Estimation using Semi-Unbalanced Neural Optimal Transport
Milena Gazdieva[*], **Jaemoo Choi**[*], Alexander Kolesov, Jaewoong Choi, Petr Mokrov, Alexander Korotin

★ Improving Neural Optimal Transport via Displacement Interpolation
Jaemoo Choi, Yongxin Chen, Jaewoong Choi

ICML 2024
(1 paper)

★ Scalable Wasserstein Gradient Flow for Generative Modeling through Unbalanced Optimal Transport
Jaemoo Choi[*], Jaewoong Choi[*] and Myungjoo Kang

ICLR 2024
(1 paper)

★ Analyzing and Improving OT-based Adversarial Networks
Jaemoo Choi[*], Jaewoong Choi[*] and Myungjoo Kang

NeurIPS 2023
(1 paper)

★ Generative Modeling through the Semi-dual Formulation of Unbalanced Optimal Transport
Jaemoo Choi[*], Jaewoong Choi[*] and Myungjoo Kang

ICML 2023
(1 paper)

★ Restoration based Generative Models
Jaemoo Choi[*], Yesom Park[*] and Myungjoo Kang

Preprints

Preprint 2026
(8 papers)

★ Scaling Reinforcement Learning for Diffusion Models via Velocity Matching
Jaemoo Choi, Wei Guo, Yuchen Zhu, Arash Vahdat, Molei Tao, Julius Berner, Yongxin Chen

★ In-Distribution Forcing for Long Video Generation at Test Time
Jeongwoo Shin[*], Youngyoon Choi[*], Sangwoo Jo, Hyunmog Kim, Sungjoon Choi[†], Joonseok Lee[†], Jaewoong Choi[†], **Jaemoo Choi**[†]

• LongTake: Learning to Sustain Dynamics in Long-Horizon Video Generation
Byoungwoo Park, **Jaemoo Choi**, Juho Lee[†], Yongxin Chen[†]

★ MetaDMD: Stable and Scalable Training of Posterior Samplers for Test-Time Steering
Jeongwoo Shin[*], Dongsoo Shin[*], Seohyun Ye, Sungjoon Choi[†], Joonseok Lee[†], Jaewoong Choi[†], **Jaemoo Choi**[†]

• Multi-Draft On-Policy Distillation for Diffusion Drafters
Petr Molodyk, Yuchen Zhu, **Jaemoo Choi**, Molei Tao, Yongxin Chen

• Hard-Constrained Sampling on Embedded Riemannian Manifolds via Adjoint Schrödinger Bridges
Mattia Mosso, **Jaemoo Choi**, Heng Yang

• Distributionally Robust Schrödinger Bridge
Jinhwan Sul, Panagiotis Theodoropoulos, Vincent Pacelli, **Jaemoo Choi**, Evangelos Theodorou

• RMA: an Agentic System for Research-Level Mathematical Problems
Zelin Zhao, Bo Yuan, **Jaemoo Choi**, Yongxin Chen

Preprint 2025
(1 paper)

★ MFM-point: Multi-scale Flow Matching for Point Cloud Generation
Petr Molodyk[*], **Jaemoo Choi**[*], David W. Romero, Ming-Yu Liu, Yongxin Chen

Invited Talks

08/2026	Distributional Transport for Generalizable and Adaptive AI KAIST Kim Jaechul Graduate School
08/2026	Reward-based Fine-tuning and Distillation: Zero- and First-order Approaches CAU-AI Core Technology Seminar, Chungang University
07/2026	Optimal Transport and Schrodinger Bridge Theory and it's Application Post-ICML Seminar, UNIST
03/2026	RL-based Fine-tuning Diffusion Models for Alignment Guest Lecture for Special Topics, Georgia Tech
02/2025	Efficient and Scalable Samplers with Optimal Transport Perspective Center for AI Natural Sciences, KIAS
03/2025	Unbalanced Optimal Transport and its Application to Estimate Wasserstein Gradient Flow and the Robust Barycenter Distribution Optimal Transport Guest Lecture, Georgia Tech
11/2024	Algorithms for Optimal Transport, the Barycenter Problem, and Their Applications Level Set Seminar, UCLA
10/2024	Recent Works on Inverse Problem via Diffusion Models FLAIR Seminar, Georgia Tech
07/2024	Various Formulations of Optimal Transport Problems and Its Application to Generative Modeling Stanford Research Institute (SRI)
03/2024	Various Formulations of Optimal Transport Problems and Its Application to Generative Modeling Center for AI Natural Sciences, KIAS
02/2024	Generative Modeling through the Semi-dual Formulation of Unbalanced Optimal Transport Invited talk for Research laboratory performance presentation, University of Seoul
02/2024	Analyzing Optimal Transport-based Adversarial Algorithms Seoul AI Hub Conference, AI Seoul
01/2024	Bridging Two Distributions through Optimal Transport 2024 Winter Workshop on PDE and Applied Mathematics, KAIST
12/2023	Generative Modeling through the Semi-dual Formulation of Unbalanced Optimal Transport Medical Imaging AI Leading Innovation Center, SNU
08/2023	Distributional Matching between Wafer Datasets Samsung Electronics Science Project Worshop, Samsung Electronics
03/2023	Feature Extraction from Wafer Map Datasets Samsung Electronics Science Project Workshop, Samsung Electronics

Teaching & Assistance Experience

Teaching Assistant	Calculus I, II, SNU	Spring 2018, Fall 2018, Fall 2019
	Mathematics for Life Sciences I, II, SNU	Fall 2020, Fall 2022, Spring 2023
Course Assistant	Optimal Transport, Georgia Tech	Spring 2025
	Engineering Mathematics, SNU	Spring 2020
	Mathematical Analysis, SNU	Fall 2021, Spring 2022
	Computer Application for Scientific Computation, SNU	Spring 2021

Academic Services

Reviewer NeurIPS (2024–present), ICLR/ICML (2025–present), CVPR (2026), Automatica (2024), TMLR (2024–2025)

Certificates

02/2018 Secondary (Middle/High) School Teacher Certificate